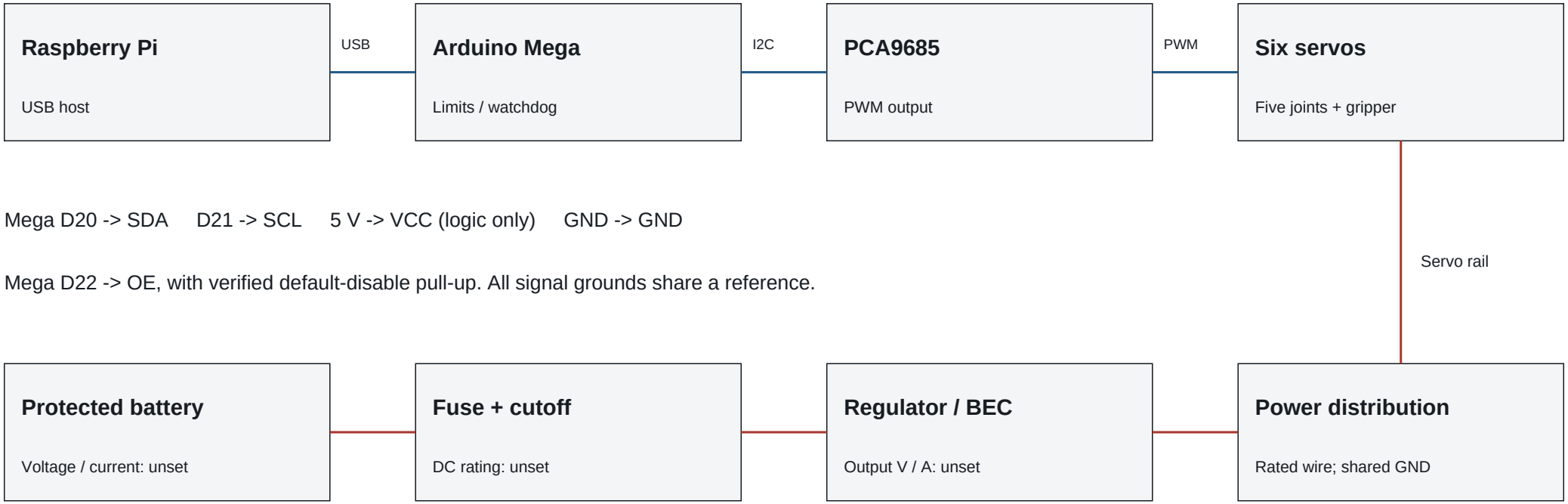
RoboLLM / Elexhub - wiring and power

DESIGN PENDING COMPONENT RATINGS - complete the worksheet before connecting servo power.



Mega D20 -> SDA D21 -> SCL 5 V -> VCC (logic only) GND -> GND

Mega D22 -> OE, with verified default-disable pull-up. All signal grounds share a reference.

1. Never power servo motors from Mega/Pi 5 V. Check the breakout's terminal and trace ratings.

2. OE disables PWM; it does not disconnect motor power. Use the physical DC-rated cutoff.

3. Check regulator input range against the fully charged AND discharged battery voltages.

4. Calibrate one unloaded servo at a time. Confirm polarity and signal pin labels on your actual board.

Sources: nxp.com/docs/en/data-sheet/PCA9685.pdf | learn.adafruit.com/16-channel-pwm-servo-driver/hooking-it-up

RoboLLM / Elexhub - measurement record

Date / operator: _____ Photo / drawing reference: _____

Battery cells / capacity / min-full V: _____

BMS / discharge rating / charger: _____

Regulator model / input V / output V-A: _____

Fuse / wire / cutoff ratings: _____

Arm base / camera / mat position: _____

Actuator	Servo model	Link mm	Axis	Limits deg	Pulse us	Mass g
Base						
Shoulder						
Elbow						
Wrist pitch						
Wrist roll						
Gripper						

Also record: neutral, direction, safe speed, voltage range, stall current, gripper opening, cutoff and watchdog evidence.

All supplied CAD dimensions are PROVISIONAL until replaced with this arm's measurements.